BSc (Hons) in Computer Science via GDSE

Module Code ITS1066

Professional Software Project in IT

CinnaCraftPro

Nuwindu Jaliya

252762002

76

(20/11/2025)

Content

No	Title	Page
1.0	Introduction to business	3
2.0	Overview of the proposed solution	4
3.0	Advantages of the solution	5
4.0	Use case diagram	6
5.0	Entity relationship diagram (ERD)	7

List of Figures

No	Title	Page
Figure 1.1	EmployeeHierarchy	**12**

List of Abbreviation

IT - Information Technology

SE - Software Engineer

QAE - Quality Assurance Engineer

SD- Software Developer

BCS - British Computer Society

HR- Human Resource

Introduction to Business: The Magedara Cinnamon Factory stands as a cornerstone of the Magedara village, providing a significant source of livelihood for the local community. With a substantial workforce primarily composed of villagers, the factory not only offers employment opportunities but also relies on local cinnamon suppliers for its raw materials. The factory plays a vital role in the village's economy, with its cinnamon products being sold locally and exported to other countries, contributing to the region's economic growth.

1)Cinnamon stick processing

Harvesting:Skilled workers meticulously select mature cinnamon quills from the cinnamon trees. The timing of harvesting is crucial to ensure the highest quality cinnamon sticks.

Trimming and Sorting: The harvested cinnamon quills undergo trimming to remove any impurities or irregularities.

Workers carefully sort the cinnamon quills based on size, thickness, and quality standards to prepare them for further processing.

Drying: The sorted cinnamon quills are then subjected to a controlled drying process. They are spread out in well-ventilated areas or drying chambers to reduce moisture content, making them more pliable for subsequent processing steps.

Cutting and Shaping: Once adequately dried, the cinnamon quills are cut into uniform lengths and shaped into sticks using specialized equipment.
Skilled craftsmen may also hand-shape the cinnamon sticks to achieve desired aesthetics and consistency.

Curing and Fermentation: In some cases, the cinnamon sticks undergo a curing or fermentation process to enhance flavor and aroma.
This step involves carefully monitored conditions to allow for natural enzymatic reactions, resulting in the development of unique flavor profiles.

Drying (Again): Following curing or fermentation, the cinnamon sticks undergo another round of drying to remove any residual moisture and stabilize their composition. Proper drying ensures the longevity and quality of the cinnamon sticks during storage and transportation.

Grading and Packaging: The processed cinnamon sticks are meticulously graded based on size, color, and aroma.
They are then carefully packaged in various forms, including bundles, bags, or containers, ready for distribution and sale to customers worldwide.

2. Consumer Sales:

- Magedara Cinnamon Factory maintains a direct relationship with its customers.To a diverse clientele that includes locals and large distributors. ● Local residents: The factory serves the immediate community by providing themWith access to fresh, locally produced cinnamon products. These sales contribute to localityBuilding a strong bond between the economy and the factory and the villagers.● Large distributors: The factory cooperates with large distributorsCinnamon products to a wider market including other regions and even internationalCustomers. This expands Magedara Cinnamon to a global audience.

Overview of the Proposed Solution:

In response to the manual and inefficient operational processes at the Magedara Cinnamon Factory, we propose the implementation of a comprehensive software solution to streamline operations and enhance efficiency:

1.Supplier Payment Tracking:

- Automate supplier payment processes to ensure accurate and timely transactions.

- Integrate automatic alerts and notifications to inform suppliers of payment status, fostering transparency and trust.

2. Worker Overtime Salary Calculation:
   - Implement a feature for precise and automated calculation of workers' overtime salaries based on recorded working hours.
   - Streamline payroll management to ensure fair compensation and efficiency.

3. Attendance Monitoring:
   - Introduce a reliable attendance tracking system to provide real-time data for improved workforce management.
   - Enhance payroll accuracy and facilitate data-driven decision-making for managers.

4. Cinnamon Sales Tracking
   - Develop a system to track all cinnamon sales orders in real-time, providing valuable insights into business performance.
   - Enable quick response to market changes and informed decisions regarding production and sales strategies.

5. Fertilizer Arrival Notification System:

   - Implement a solution to notify cinnamon suppliers when fertilizer arrivals are available.
   - Improve communication with suppliers, optimize production processes, and ensure a steady supply of fertilizers.

Advantages of the Solution:

1. Improved Supplier Payment Tracking:
   - Automation eliminates errors, ensures timely payments, and enhances transparency in supplier relationships.
   - Reduction in manual effort leads to increased operational efficiency and cost savings.

2. Enhanced Worker Overtime Salary Calculation:

- Precise calculations minimize errors, ensure fair compensation, and boost employee satisfaction.
- Streamlined payroll management saves time and resources.

3. Accurate Attendance Monitoring:

- Real-time tracking provides accurate data for improved workforce management and payroll calculations.
- Data-driven decisions based on real-time attendance information enhance operational efficiency.

4. Real-Time Cinnamon Sales Tracking:

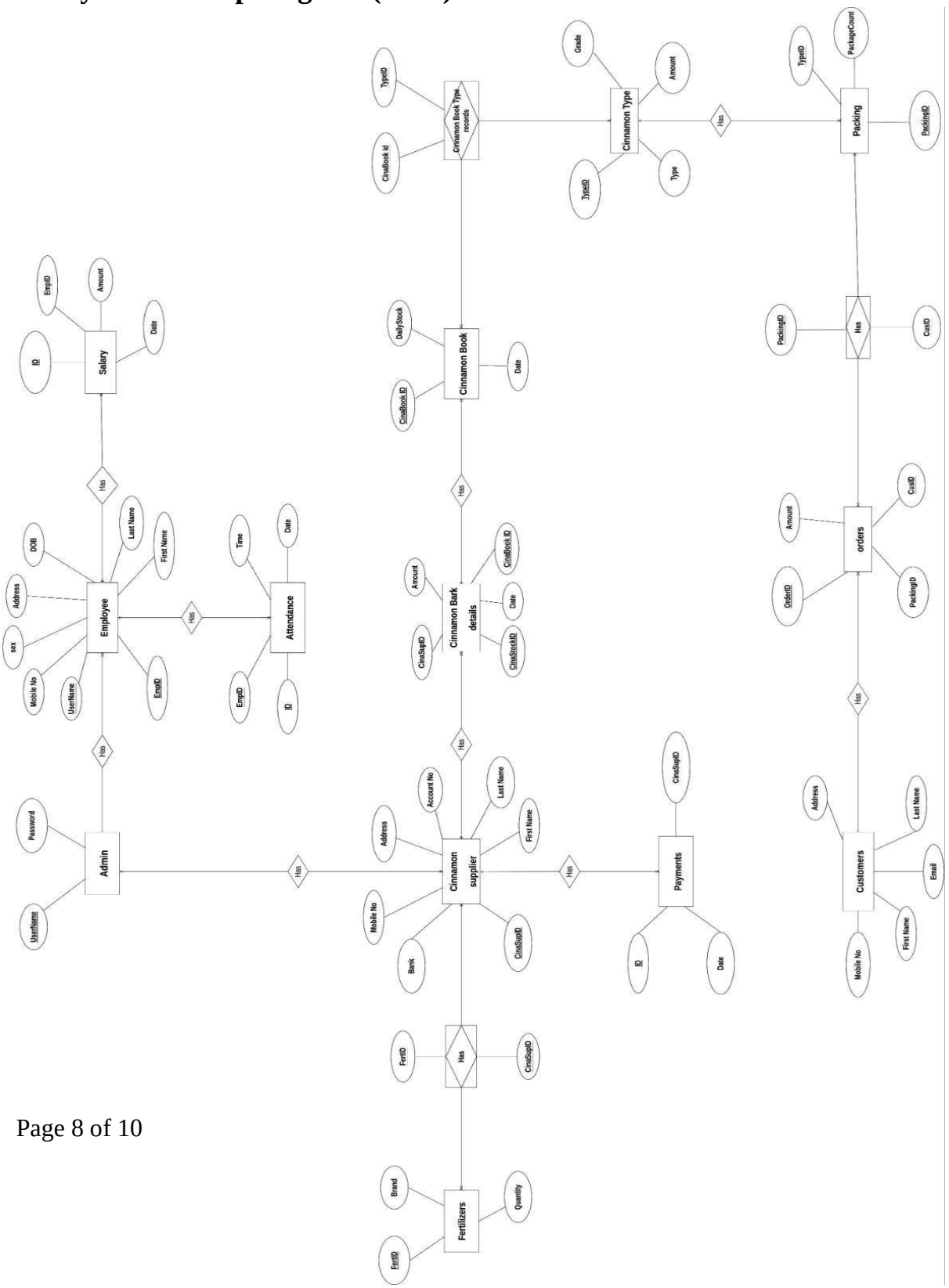
- Insights into business performance enable quick responses to market changes and better inventory management.
- Enhanced demand forecasting facilitates improved production and sales strategies.

5. Efficient Fertilizer Arrival Notifications:

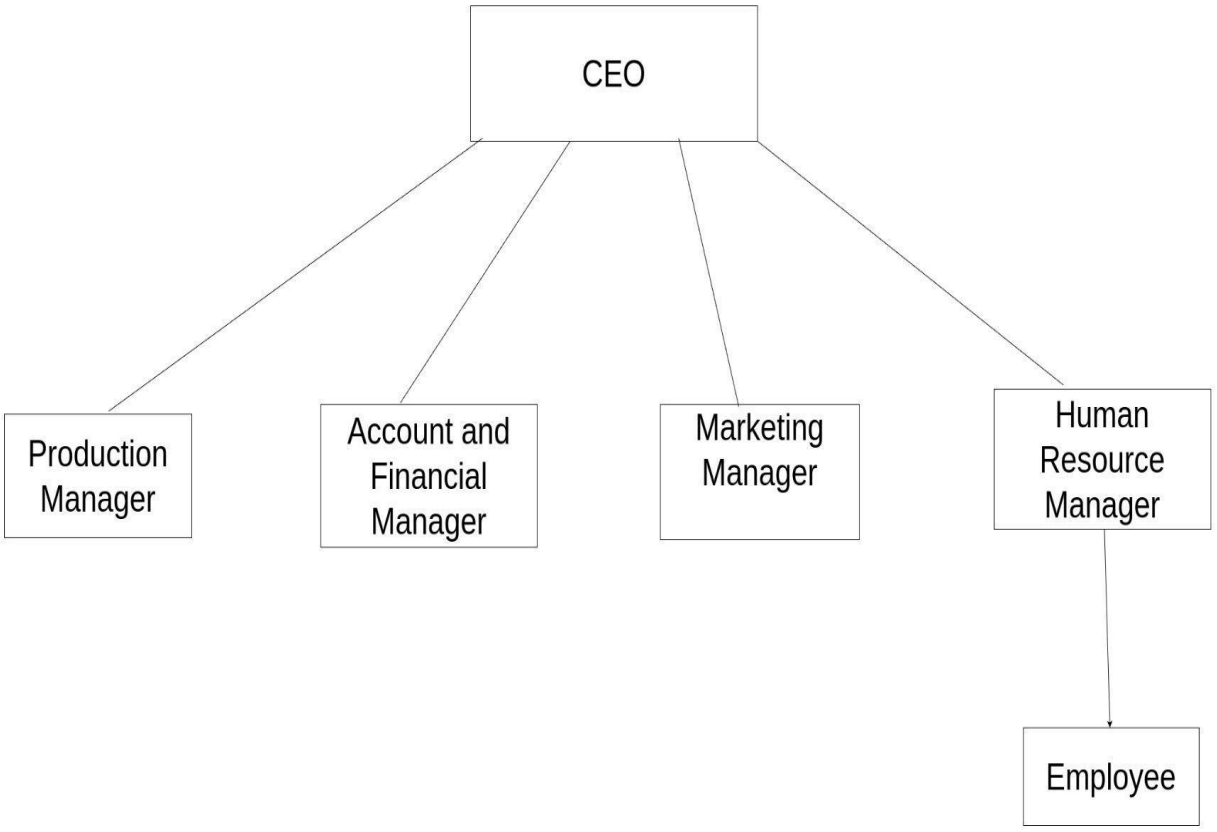
- Improved communication with suppliers optimizes production processes and ensures a steady supply of fertilizers.
- Reduction in delays and disruptions in the supply chain leads to improved efficiency and productivity.

Use case diagram

Entity relationship diagram (ERD)



Employee Hierarchy



References - – The following are just examples for your convenience

https://www.srilankabusiness.com/spices/about/cinnamon-cultivation-sri-lanka.html

https://youtu.be/oowuwUKkxjs?si=cmyz3Q9hHatyvPWd

https://youtu.be/OhO5fKkHTH8?si=0uIyIbnfDKnm-Pzb

https://www.slofoodgroup.com/products/ceylon-cinnamon-sticks